# Online Retail Sales & Customer Analytics

**Project Proposal**

## 1. Project

**RetailIQ: Online Retail Sales & Customer Analytics**

## 2. Objective

To analyze online retail sales patterns, identify the main operational and behavioral drivers of sales performance, and develop forecasting models that predict sales for the next operating period. The project will also estimate the expected order volume and revenue based on the predicted sales, helping decision makers at UCI, a UK-based online retail company, improve inventory planning, marketing strategy, and customer relationship management.

## 3. Description

Online retail performance can be significantly affected by seasonality, product demand, customer behavior, returns, and market conditions across different countries. RetailIQ aims to transform raw e-commerce transaction data into actionable insights through data modeling, data cleaning, preprocessing, exploratory analysis, statistical investigation, forecasting, and interactive visualization.

The project will investigate sales trends across products, customers, countries, invoices, order timing, returns, and other operational conditions, depending on the available dataset. It will identify important factors contributing to sales performance and customer value, and develop forecasting solutions, particularly for predicting expected sales on the next operating day.

Based on the predicted sales, the project will estimate expected revenue and highlight the expected number of orders and units that can be fulfilled. The final revenue and order-capacity calculation will depend on the fields available in the dataset and clearly stated assumptions.

## 4. Tools & Design

The project will follow an end-to-end data analytics workflow:

- Data understanding and data modeling.
- Data cleaning and preprocessing.
- Exploratory and diagnostic analytics.
- Forecasting and predictive analytics.
- Interactive Power BI dashboard development and business communication.

## 5. Technologies to Use

- SQL: Data modeling, extraction, transformation, cleaning, aggregation, and analytical queries.
- Python: Data preprocessing, exploratory analysis, statistical analysis, forecasting, and model evaluation.
- Python libraries: pandas, NumPy, Matplotlib, and scikit-learn.

- Power BI: Interactive dashboard, KPI reporting, trend analysis, forecasting visualization, and business storytelling.
- DAX: Measures, calculated metrics, KPI logic, and analytical calculations within Power BI.
- Microsoft Excel: Optional supporting tool for quick validation and data inspection.

## 6. Team Members & Responsibilities

| Name | Responsibilities |
|---|---|
| Anas Gaber | Data model, SQL structure, SQL queries, data validation, and DAX support. |
| Yamen Amr | Data cleaning, preprocessing, data quality checks, and documentation. |
| Asmaa Mokhtar | Analysis questions, KPI calculations, exploratory analysis, and dashboard planning. |
| Hassan Alaa | Sales-driver and customer-behavior analysis, business insights, customer-value analysis, and recommendations. |
| Yasmin Mahmoud | Forecasting, model evaluation, next-period sales prediction, Power BI dashboard development, and presentation coordination. |

## 7. Week 1: Data Model, Data Cleaning, and Preprocessing

### *Tasks*

- Understand the dataset structure, business context, and meaning of each variable (invoices, products, quantities, prices, customers, and countries). **Responsible: All team members.**
- Identify the main entities, dimensions, measures, and relationships within the dataset. **Responsible: Anas Gaber.**
- Design the data model and determine the appropriate analytical tables. **Responsible: Anas Gaber.**
- Inspect data types, missing CustomerID values, duplicate records, cancelled orders, negative quantities, inconsistent country names, and potential outliers in price and quantity. **Responsible: Yamen Amr.**
- Clean and preprocess the data using SQL and Python. **Responsible: Anas Gaber and Yamen Amr.**
- Create a documented preprocessing notebook explaining cleaning decisions and transformations. **Responsible: Yamen Amr.**
- Validate the cleaned data and confirm that it is suitable for analysis and forecasting. **Responsible: Anas Gaber and Yasmin Mahmoud.**

### *Deliverables*

- Documented data model.
- Cleaned and analysis-ready dataset.
- SQL scripts for data extraction, transformation, and cleaning.
- Data preprocessing notebook.
- Data dictionary describing important fields.
- Data quality summary and validation results.

## 8. Week 2: Analysis Questions and Exploratory Analytics

### *Tasks*

- Determine the main business questions that can be answered using the available dataset. **Responsible: Asmaa Mokhtar, with input from all team members.**
- Identify important operational KPIs, such as total revenue, average order value, order frequency, return rate, and customer lifetime value, depending on available fields.
- Analyze sales trends by day, week, month, product, customer, and country, where applicable.
- Identify products, customers, and countries associated with the highest and lowest sales performance.
- Investigate relationships between order volume, product demand, customer segments, and returns.
- Use SQL and Python to calculate metrics, aggregate data, and generate visualizations. **Responsible: All team members.**
- Prioritize meaningful findings and convert them into business-oriented insights. **Responsible: Asmaa Mokhtar and Hassan Alaa.**

### *Deliverables*

- Documented list of analysis questions.
- SQL queries answering the selected questions.
- Exploratory data analysis notebook.
- KPI calculations and supporting visualizations.
- Analysis of major sales patterns and customer behavior.
- Initial business insights and recommendations.
- Shortlist of variables to be used in forecasting.

## 9. Week 3: Forecasting and Predictive Analytics

### *Tasks*

- Determine forecasting questions based on the dataset and findings from Week 2.
- Develop a forecasting approach for predicting sales on the next operating day. **Responsible: Yasmin Mahmoud.**
- Prepare time-based features and relevant explanatory variables for forecasting.
- Compare suitable forecasting or machine-learning approaches, depending on dataset structure and available historical records.
- Evaluate model performance using appropriate metrics and validation methods.
- Estimate expected revenue and order volume based on the predicted sales.
- Calculate or highlight the expected number of orders and units that can be fulfilled on the next operating day, based on available operational information and clearly stated assumptions.
- Create visualizations showing actual versus predicted sales and expected revenue.

### *Deliverables*

- Documented forecasting questions.
- Forecasting methodology and feature-selection explanation.
- Trained forecasting or predictive model.
- Model evaluation results.
- Next-day sales prediction.
- Estimated revenue and expected order/unit volume.
- Actual-versus-predicted visualizations.
- Forecasting notebook and documented assumptions.
- Limitations and recommendations for improving the model.

## 10. Week 4: Power BI Dashboard, Final Report, and Presentation

### *Tasks*

- Design the overall dashboard structure and determine the most important pages and visualizations.
- Build an interactive Power BI dashboard containing KPIs, sales trends, breakdowns, customer segments, product performance, and forecasting results.
- Create a Power BI data model and DAX measures required for KPI calculations and dashboard analysis.
- Add filters and interactive controls for relevant dimensions such as date, country, product, and customer, where available.
- Integrate the final analytical and forecasting outputs into the Power BI report.
- Create a management-focused view showing the main sales drivers, expected next-day sales, and estimated revenue.
- Validate dashboard calculations, visual consistency, and usability.
- Prepare the final report and presentation, including the business problem, methodology, data preparation, analysis, forecasting model, results, limitations, and recommendations.
- Rehearse the final presentation and ensure that each team member can explain their contribution.

### *Deliverables*

- Interactive Power BI dashboard.
- Power BI report and data model with DAX measures.
- Power BI dashboard documentation and KPI definitions.
- Final analytical and forecasting results.
- Final report.
- Final presentation.
- Business recommendations for increasing revenue and improving customer retention.
- Complete project files, including SQL scripts, Python notebooks, cleaned datasets, and Power BI files.